BUILD STAGE

Github and Build Submission Guidelines

1. Submission Requirements

The following submissions are expected during the Build stage.

Item	Requirement
GitHub Repository	Working source code with proper project structure
Working Application	Functional application/prototype
Database Implementation	Implemented database and connectivity
API Implementation	Implemented internal/external APIs, where applicable
Testing Evidence	Test cases and results

2. GitHub Repository – Recommended Content

The repository should contain:

- Source code
- Project structure
- README
- Installation instructions
- Requirements/dependencies
- Database scripts/schema
- API documentation, where applicable
- Sample/test data, where required

README should clearly explain:

- Project overview
- Features
- Technology stack
- Installation steps
- How to run the application
- Database setup
- API setup
- Configuration/environment variables
- Usage instructions

3. Working Application

Demonstrate the actual working system. The application should demonstrate, where applicable:

- User registration/login
- Core user workflow
- Major modules
- CRUD operations
- Database interaction
- API interaction
- Validations
- Error handling
- Authentication/authorization
- AI/ML functionality
- Notifications/integrations

The Build stage should focus on **working functionality rather than only static screens**.

4. Database Implementation

Maintaining the database designed during the Design stage has been implemented.

Include:

- Tables
- Primary keys
- Foreign keys
- Relationships
- Sample data
- CRUD operations
- Database connectivity
- Important queries

5. API Implementation

Where APIs are used, provide:

- API name/endpoint
- Purpose
- Request
- Response
- Authentication
- External API details

6. Testing

Teams maintain different Test cases for testing the development.

Test Case	Expected Result	Actual Result	Status
Login with valid credentials	Login successful	Login successful	Pass
Invalid password	Error message	Error message	Pass
Submit form	Data saved	Data saved	Pass

Also mention:

- Bugs identified
- Bugs fixed
- Remaining issues
- Retesting performed

8. Build Quality Guidelines

● **Maintain code quality:** Follow a clean and understandable project structure with comment lines.

● **Use meaningful names:** Use meaningful names for files, classes, variables and functions.

● **Maintain GitHub properly:** Keep the repository organized and updated.

● **Maintain design consistency:** The implementation should follow the approved Design.

● **Protect credentials:** Never expose API keys, passwords or sensitive credentials in the repository.

9. Common Mistakes to Avoid

● Having an incomplete README.

● Hardcoding sensitive credentials. (Remove Your API keys before uploading)

● Having code that significantly differs from the approved design without explanation.